

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
PART TEST – II
PAPER –2
TEST DATE: 12-12-2021

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 – 10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

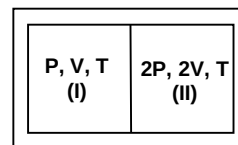
Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. A partition divides a container having insulated walls into two compartments 1 and 2. Same gas fills the two compartments and the initial temperature, pressure and volume of gas is shown in the diagram [in the left chamber pressure, volume and temperature are (P, V, T) and in the right chamber it is (2P, 2V, T)]. The partition is conducting and can move freely without friction. Which of the following statements is/are correct, with reference to final equilibrium position?

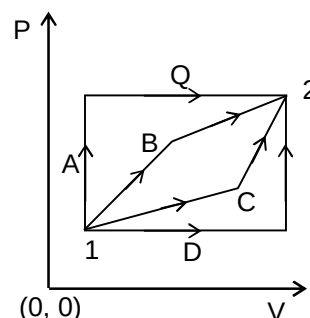


- (A) The pressure in the two compartments will be equal.
 (B) The volume of compartment 1 will be $3V/5$.
 (C) The volume of compartment 1 will be $2V/5$.
 (D) The pressure in compartment 1 will be $5P/3$

Ans. A, B, D

Sol. At equilibrium temperature and pressure on both sides will be same .

2. An ideal gas is taken from state 1 to state 2 through optional path A, B, C and D as shown on P-V diagram . Let Q, W and U represent heat supplied to gas, work done by gas and change in internal energy of the gas respectively, then:



- (A) $Q_B - W_B > Q_C - W_C$
 (B) $Q_A - Q_D = W_A - W_D$
 (C) $W_A < W_B < W_C < W_D$
 (D) $Q_A > Q_B > Q_C > Q_D$

Ans. B, D

Sol. Change in internal energy of the gas is path independent and use first law of thermodynamics.

3. The nominal voltage of a light bulb is $V_0 = 26 \text{ V}$ and nominal current $I_0 = 0.12 \text{ A}$. Cold tungsten filament of this light bulb has resistance $R_0 = 24 \Omega$. L is length of the filament and d is the diameter. T is the temperature at which this filament is supposed to emit light (when working in nominal regime). The resistivity of tungsten at room temperature is $\rho_0 = 5.3 \times 10^{-8} \Omega \cdot \text{m}$. For metals, the resistivity can be taken to be proportional to the temperature in Kelvins. Treat tungsten as a perfectly grey body which absorbs $k = 0.3$ of the incident radiation at any wavelength. Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$. Neglect thermal expansion of the filament. T_0 is the room temperature.

(A) $T = \frac{V_0 T_0}{I_0 R_0}$

(B) $d = \sqrt[3]{\frac{4V_0 I_0 \rho_0}{\pi^2 R_0 k \sigma T^4}}$

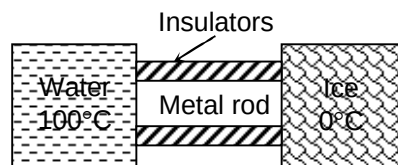
(C) $L = \frac{R_0 d^2 \pi}{4\rho_0}$

(D) $d = \sqrt[3]{\frac{4V_0 I_0 \rho_0}{\pi^2 R_0 k \sigma T_0^4}}$

Ans. A, B, C

Sol. Use Ohm's law for the tungsten resistance. The resistivity of the resistance will change as it is not being operated at room temperature.

4. An insulated container is filled with ice at 0°C , and another container is filled with water that is continuously boiling at 100°C . In series of experiments, the container is connected by various, thick metal rods that pass through the walls of the container as shown in the figure.



In experiment I : A copper rod is used and all ice melts in 20 minutes

In experiment II : A steel rod of identical dimensions is used and all ice melts in 80 minutes

In experiment III : Both rods are used in series and all ice melts in t_1 minutes.

In experiment III : Both rods are used in parallel and all ice melts in t_2 minutes.

(A) $t_1 = 100$

(B) $t_1 = 50$

(C) $t_2 = 16$

(D) $t_2 = 8$

Ans. A, C

Sol. When the rods are connected alone use $\frac{dq}{dt} = \frac{\Delta T}{\frac{\ell}{KA}}$

For series combination use : $\frac{dq}{dt} = \frac{\Delta T}{\frac{\ell_1}{K_1 A_1} + \frac{\ell_2}{K_2 A_2}}$

For parallel use : $\frac{dq}{dt} = \Delta T \left(\frac{K_1 A_1}{\ell_1} + \frac{K_2 A_2}{\ell_2} \right)$

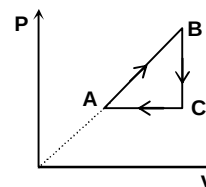
5. Three bodies A, B and C have equal surface area and thermal emissivity in the ratio $e_A : e_B : e_C = 1 : \frac{1}{2} : \frac{1}{4}$. All three bodies are radiating heat at the same rate. Their wavelengths corresponding to maximum intensity are $\lambda_A, \lambda_B, \lambda_C$ respectively and their temperatures are T_A, T_B and T_C on kelvin scale . Select the correct statement/s:

- (A) $\sqrt{T_A T_C} = T_B$
 (B) $\sqrt{\lambda_A \lambda_C} = \lambda_B$
 (C) $\sqrt{e_A e_C T_A T_C} = e_B T_B$
 (D) $\sqrt{e_A e_B T_A T_B \lambda_A \lambda_B} = e_C T_C \lambda_C$

Ans. A, B, C

Sol. Rate of emission is given by $\frac{dQ_e}{dt} = \sigma A e T_{\text{body}}^4$. Along with this use Wein's displacement law .

6. Consider a thermodynamic cycle in a PV diagram shown in the figure performed on one mole of a ideal monoatomic gas. The temperature at A is T_0 and volume at A and B are related as $V_B = V_C = 2V_A$. Choose the correct option(s) from the following.



- (A) The maximum temperature during the cycle is $4T_0$
 (B) Net work done by the gas during the cycle is $0.5 RT_0$
 (C) The heat capacity of the process AB is $2R$.
 (D) The efficiency of the cycle is 8.33%

Ans. A, B, C, D

Sol. $P_A V_A = RT_0$
 $P_B = 2P_A$
 $V_B = V_C = 2V_A$
 $T_B = \frac{P_B V_B}{R} = 4T_0$
 $\Delta W = \left(\frac{V_C - V_A}{2} \right) (P_B - P_C) = 0.5 RT_0$

$$\Delta W_{AB} = \frac{1}{2}(P_A + P_B)(V_B - V_A) = \frac{3}{2}P_A V_A = 1.5RT_0$$

$$\Delta U_{AB} = \frac{3}{2}R(T_B - T_A) = 4.5 RT_0$$

$$\Delta Q_{AB} = \Delta W_{AB} + \Delta U_{AB} = 6RT_0$$

$$C = \left(\frac{\Delta Q_{AB}}{T_B - T_A} \right) = 2R$$

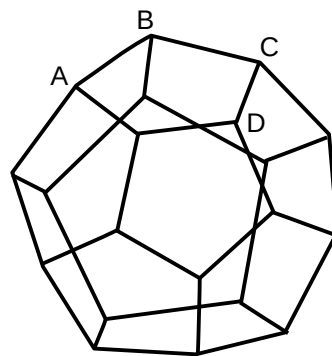
$$\eta = \frac{\Delta W_{NET}}{\Delta Q_{AB}} = \frac{1}{12} = 8.33\%$$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

A dodecahedron is shown to you in the diagram. Each edge of the dodecahedron has a resistance $R=330\Omega$. Points A, B, C and D are marked in the diagram :



7. The value of $R_{AB}:R_{CD}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Ans. A

Sol. Use symmetry arguments.

8. Find R_{AB} .

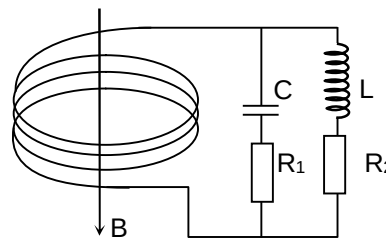
- (A) 209Ω
- (B) 570Ω
- (C) 500Ω
- (D) 770Ω

Ans. A

Sol. Use symmetry arguments to prove that the equivalent resistance is $19R/30$

Paragraph for Question Nos. 09 and 10

Consider an electric circuit consisting of a coil of negligibly small inductance, consisting of $N = 10$ turns and with the surface area of a single loop $S = 10 \text{ cm}^2$, resistors $R_1 = R_2 = 3\Omega$, capacitor $C = 0.2 \text{ F}$, and an inductance $L = 1\text{H}$, connected as shown in Fig. At the moment of time $t = 0$, a magnetic field, parallel to the axis of the coil is switched on. The induction of the magnetic field starts growing linearly, starting from $B = 0$ until the maximal value $B = 1\text{T}$ is achieved at $t = 10 \text{ ms}$. Further, the induction of the magnetic field remains constant (and equal to 1T).



9. Find the current through the resistors R_2 at the moment of time $t_1 = 5 \text{ ms}$.

- (A) 5mA
- (B) 10mA
- (C) 20 mA
- (D) 40mA

Ans. A

Sol. During the linear growth of B (from 0 ms to 10 ms , the coil serves as an ideal battery of emf. $NS(dB/dt)$. Estimate the characteristic times for two current loops: first involving the coil and C , and second, involving the coil and L ; compare this with the time-scale of 10 ms .

10. What is the net charge passing through the resistor R_2 ?

- (A) 1.5mC
- (B) 3.3mC
- (C) 7.6mC
- (D) 13.3mC

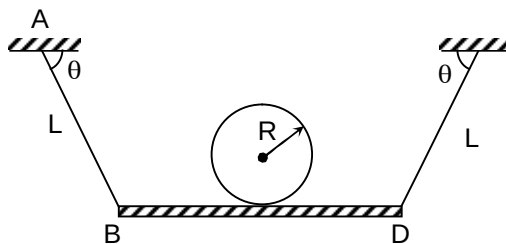
Ans. B

Sol. Apply KVL for the current loop consisting of the coil, L , and R_2 .

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A metal cylinder of radius R is placed on a wooden plank BD . The plank is kept horizontal suspended with the help of two identical string AB and CD each of length L . The temperature coefficient of linear expansion of the cylinder and the strings are α_1 and α_2 respectively. Angle θ shown in the figure is 30° . It was found that with change in temperature the centre of the cylinder did not move. Find the ratio $\frac{\alpha_1}{\alpha_2}$, if it is known that $L = 4R$. Assume that change in value of θ is negligible for small temperature changes.



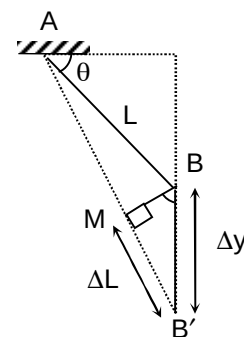
Ans. 8

- Sol. Let change in temperature be ΔT
 Length of a string changes by $\Delta L = L\alpha_2\Delta T$
 The wooden plank descends by $\Delta y = \frac{\Delta L}{\sin\theta}$

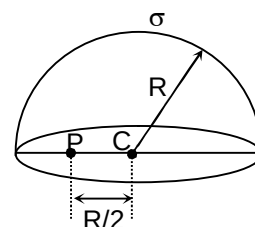
$$\Delta y = 2\Delta L \left[\because \sin\theta = \frac{1}{2} \right]$$

 Change in radius of the ball : $\Delta R = R\alpha_1\Delta T$
 The centre of the ball will not move if $\Delta y = \Delta R$

$$\Rightarrow 2L\alpha_2\Delta T = R\alpha_1\Delta T \Rightarrow 8R\alpha_2 = R\alpha_1 \Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{8}{1}$$



12. Consider a uniformly charged hemispherical shell having charge per unit area σ . Find the potential (in Volts) at point P due to the shell. R is the radius of shell. The distance of point P from the center is $R/2$ (see diagram). given: $\frac{\sigma R}{\epsilon_0} = 4$ SI units.



Ans. 2

- Sol. The field from P to C will be perpendicular to line PC . Hence, potential at P = Potential at C .
13. An electric bulb of $100 \text{ W} - 300 \text{ V}$ is connected with an A.C. supply of 500 V and $150/\pi \text{ Hz}$. The required inductance to save the electric bulb is $\lambda \text{ H}$. Find the value of λ .

Ans. 4

- Sol. $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{300^2}{100} = 900 \Omega$

$$\therefore \text{current } i = \frac{V}{R} = \frac{300}{900} = \frac{1}{3} \text{ A}$$

Let L be the required inductance, then

$$\frac{500}{Z} = \frac{1}{3} \text{ or } \frac{500}{\sqrt{(900)^2 + X_L^2}} = \frac{1}{3}$$

$$X_L = 1200 \Omega$$

$$L = \frac{X_L}{\omega} = \frac{1200}{2\pi f} = \frac{1200}{2\pi \times \frac{150}{\pi}} = 4 \text{ H}$$

Section – C (Maximum Marks: 12)

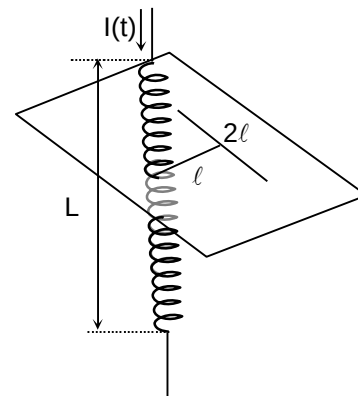
This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

The number of turns of a solenoid (straight coil) of length $L=0.5$ m is $N=2000$, the area of its cross section is $A=16$ cm², and it has an iron core of relative permittivity of 2500. At a moment the current in the coil changes at a rate of 100 A/s. What is the induced voltage (in volts) in that long straight wire of length $2l = 20$ cm (the wire is next to the centre of the coil)

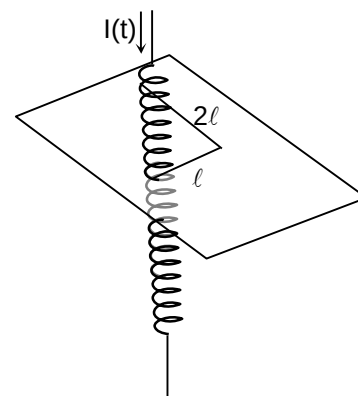
14. Which is perpendicular to the axis of the coil, if its two ends are equidistant from the coil, and its centre is at a distance of l from the coil?



Ans. 0.50

Sol. The voltage induced will be: $\frac{\mu_0 \mu_r N A \Delta I}{4L \Delta t} = 0.5 \text{ V}$

15. Which has the same direction as the previous one, and one of its end is at a distance of l from the axis of the coil?



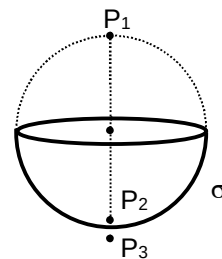
Ans. 0.35

Sol. The voltage induced will be : $\frac{\mu_0 \mu_r N A \Delta I}{\pi L \Delta t} = 0.35V$

Question Stem for Question Nos. 16 and 17

Question Stem

Consider a uniformly charged hemispherical shell having charge per unit area σ . E_1 is the electric field at P_1 (which is at a distance R from the centre of the hemispherical shell and lying on an imaginary sphere having same radius as the given hemisphere (see adjacent diagram), E_2 is the electric field at P_2 (which is a point just inside the hemispherical shell) and E_3 is the electric field at P_3 (which is a point just outside the hemispherical shell).



16. The value of $E_1:E_2$ is

Ans. 1.00

Sol. If you consider the whole sphere the net field at P_1 will be zero. Now, make use of symmetry arguments.

17. The value of $E_2:E_3$ is

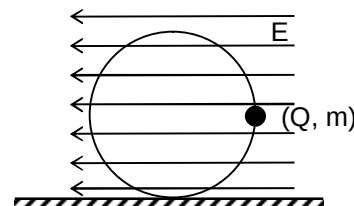
Ans. 0.17

Sol. The ratio of fields will come out to be equal to $\frac{2-\sqrt{2}}{2+\sqrt{2}}$

Question Stem for Question Nos. 18 and 19

Question Stem

A charge particle of charge q and mass m is clamped rigidly at the circumference of a ring of mass m and radius R . Initially ring is in vertical plane resting on a sufficiently rough horizontal surface, with charge q at the same horizontal level as the center of the ring. There exists a uniform horizontal electric field E as shown in the figure. At $t = 0$ the system is set free. (Given that $qE = mg$)



18. The magnitude of frictional force acting on the ring immediately after the ring is released is.....

Ans. 0.00

Sol. Initial torque about point of contact is zero. Hence the initial angular acceleration of the ring will be zero. Now, consider torque about the centre of the ring to prove that the initial frictional force on the ring is zero.

19. The work done (in Joules) by electric field during the time ring rotates by 90° is
(Take $mgR = 7\text{Joules}$ and $\pi = \frac{22}{7}$)

Ans. 7.00

Sol. Electric field will be exerting a constant force on the particle hence work done will be $QER = mgR = 7$.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. Identify the correct option(s)

- (A) Diastereomers can be separated by fractional distillation
- (B) Enantiomers are also called as optical antipodes
- (C) 'Epimers' and 'Anomer's are the type of diastereomers
- (D) Geometrical isomers are diastereomers

Ans. A, B, C, D

Sol. Enantiomers are also called as optical antipodes.

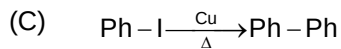
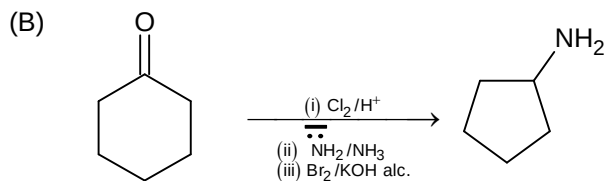
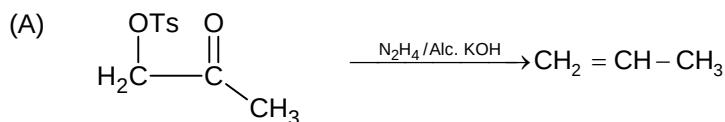
21. Correct statement(s) is/are

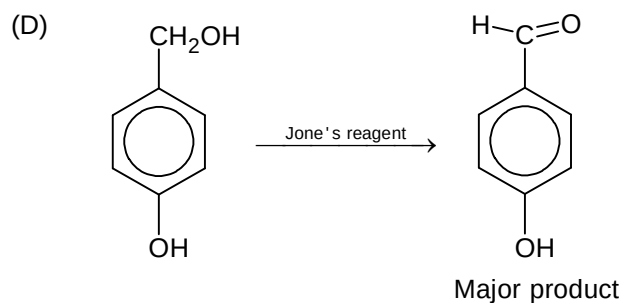
- (A) Benzyl alcohol also gives Lucas test as fast as 3° alcohol
- (B) Beilstein test is used to detect the presence of chlorine
- (C) $(\text{CH}_3)_2\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ has more enol content than $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$
- (D) RCH_2OH gives reaction with Heinsberg reagent

Ans. A, B, C, D

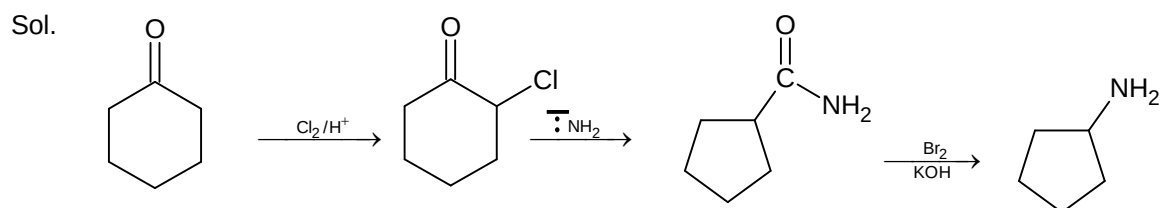
Sol. Benzyl cation is stable. Enol of $(\text{CH}_3)_2\text{CHCOCH}_3$ is more stable.

22. Which of the following reaction(s) and product(s) are correct?

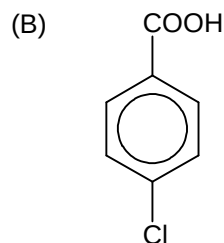
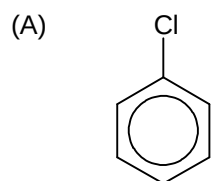




Ans. A, B, C, D



23. Which of the following compounds will respond to the Beilstein test positively?



Ans. A, B, D

Sol. Urea also gives positive test.

24. Which of the following used to detect protein?

(A) Biuret test

(B) Molisch test

(C) Xanthoprotic test

(D) Ninhydrin test

Ans. A, C, D

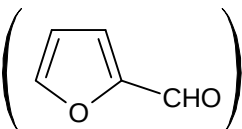
Sol. All are the tests of protein except Molisch test which is for carbohydrate.

25. The Cannizzaro reaction is given by

(A) Benzaldehyde

(B) Formaldehyde

(C) Glyoxal

(D) Furfural 

Ans. A, B, C, D

Sol. All can give Cannizzaro reaction.

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

Aldehydes and Ketones are very reactive towards nucleophilic addition reaction because they have polar

 $\text{C}=\text{O}$ group. Polarity arises due to electronegativity difference between carbon and oxygen.

Reactivity of carbonyl compound towards nucleophilic addition increases with the electron deficiency at carbonyl carbon.

26. $\text{CH}_3 - \text{CHO}$ shows nucleophilic addition reaction with

(A) $\text{CHCl}_3 / \ddot{\text{O}}\text{H}^-$

(B) NaHSO_3

(C) N_2H_4

(D) All of these

Ans. D

Sol. All of these gives nucleophilic addition reaction with $\text{CH}_3 - \text{CHO}$.

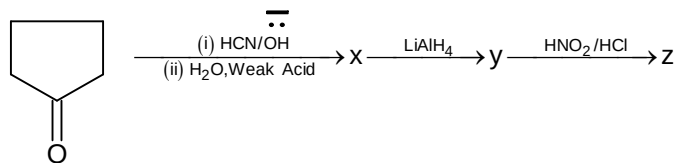
27. Which of the following reaction(s) does not involve in nucleophilic attack on $\text{C}=\text{O}$ group?

- (A) Aldol reaction
- (B) Cannizzaro reaction
- (C) Perkin reaction
- (D) Rosenmund's reduction

Ans. D

Sol. In Rosenmund reduction it's catalytic reduction of RCOCl into RCHO .

Paragraph for Question Nos. 28 and 29



28. $\text{Z} \xrightarrow[\text{(iii) H}_2\text{O}/\Delta]{\text{(i) NH}_2\text{OH}, \text{(ii) H}^+} \text{W}$, W is

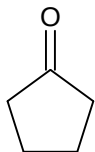
- (A)
- (B)
- (C)
- (D) Nylon-6

Ans. D

29. $\text{Z} \xrightarrow[\text{(ii) H}^+]{\text{(i) Alkaline KMnO}_4/\Delta} \text{N} \xrightarrow[\text{Excess}]{\text{C}_2\text{H}_5\text{OH/conc. H}_2\text{SO}_4} \text{P}$,

- (A) P is $(\text{CH}_2)_4(\text{COOC}_2\text{H}_5)_2$
- (B) P on reaction with $\text{C}_2\text{H}_5\text{O}^-\text{Na}^+$ gives β -keto ester (Q)

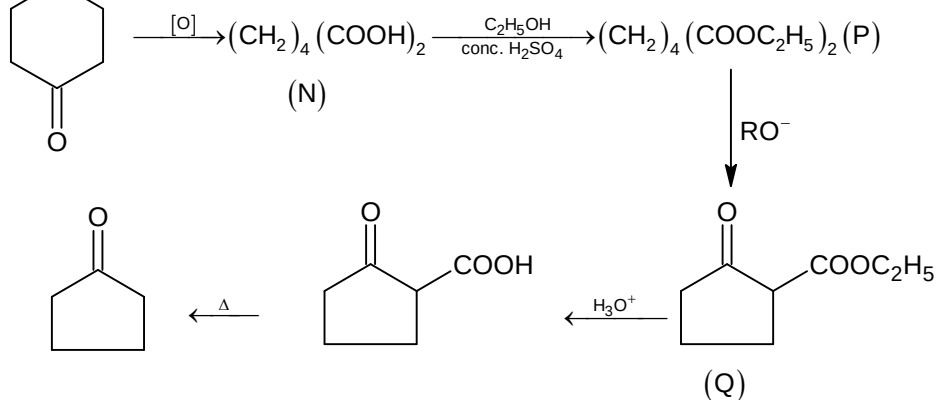
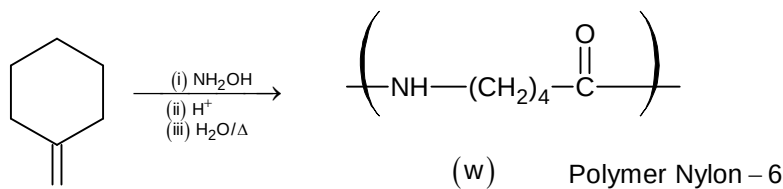
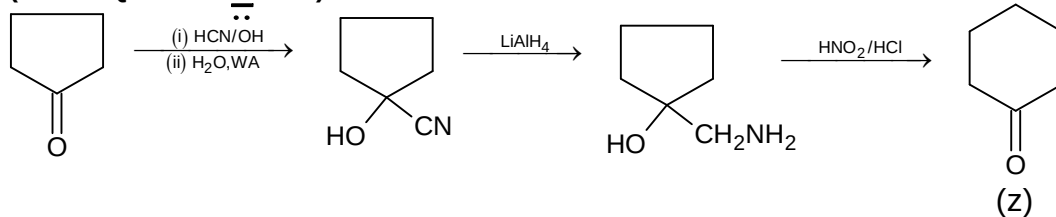
(C) Q on heating after hydrolysis gives



(D) All of these

Ans. D

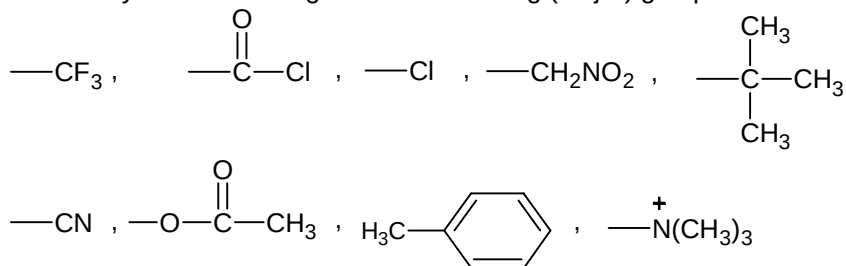
Sol. (for the Q. No. 28 and 29)



Section – B (Maximum Marks: 12)

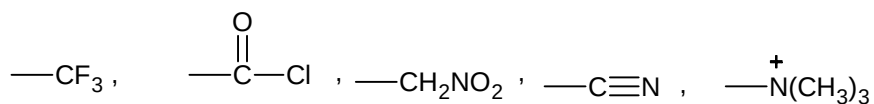
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. How many of the following are meta directing (Major) groups

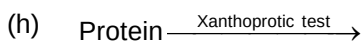
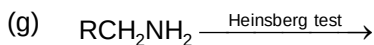
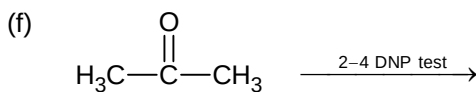
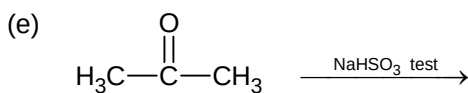
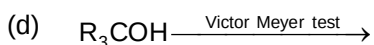
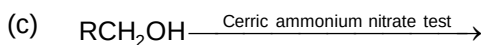
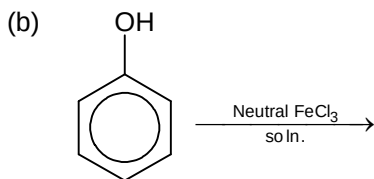
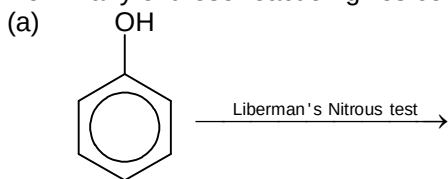


Ans. 5

Sol.



31. How many of these reaction gives coloured product (s) (not white)



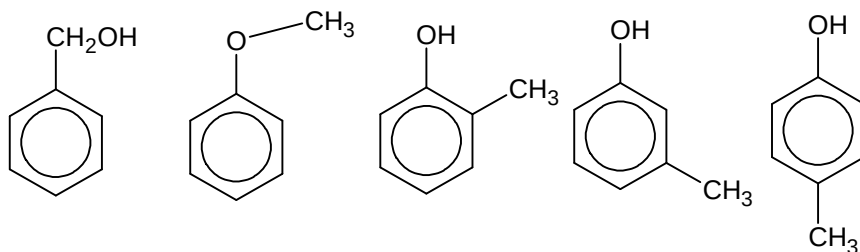
Ans. 5

Sol. (d), (e) and (g) are colourless/white.

 32. Number of isomers with the formula $\text{C}_7\text{H}_8\text{O}$ containing a benzene ring

Ans. 5

Sol.

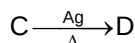
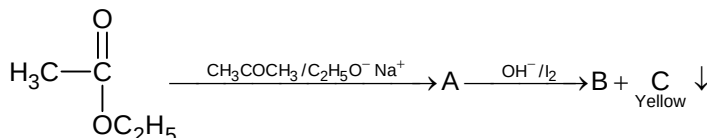


Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem



If 8.8 gram of $\text{CH}_3\text{COOC}_2\text{H}_5$ was taken, then the mass of product D is found to be x gram and mass of product A is found to be y gram. (Assume 100% efficiency of each reaction).
[Atomic mass of C = 12, H = 1, O = 16]

33. Value of x is _____

Ans. 1.30

Sol. 1 mole of $\text{CH}_3\text{COOC}_2\text{H}_5$ gives 1 mole of $\underset{(C)}{\text{CHI}_3}$

1 mole CHI_3 gives $\frac{1}{2}$ mole $\underset{(D)}{\text{CH}\equiv\text{CH}}$

$\therefore \frac{8.8}{88}$ mole of $\text{CH}_3\text{COOC}_2\text{H}_5 \rightarrow \frac{1}{2}\left(\frac{8.8}{88}\right)$ moles of C_2H_2

$W_{\text{C}_2\text{H}_2} = \frac{1}{2} \times 0.1 \times 26 = 1.30 \text{ gm}$

34. Value of y is _____

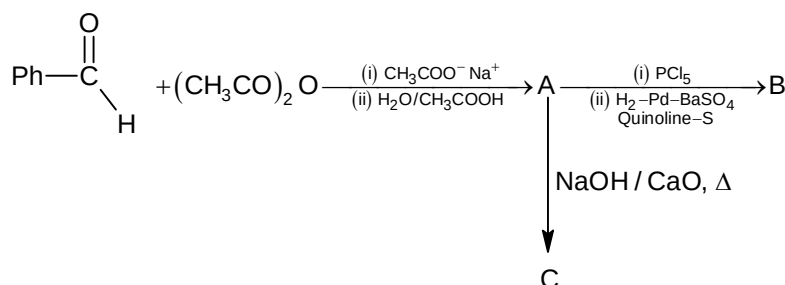
Ans. 10.00

Sol. A is $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_2-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$
Molar mass = 60 + 8 + 32 = 100

$\therefore 8.8 \text{ gm gives} = \frac{100}{88} \times 8.8 = 10.00 \text{ gm}$

Question Stem for Question Nos. 35 and 36

Question Stem



35. If molar mass of C = M then M / 5 is _____

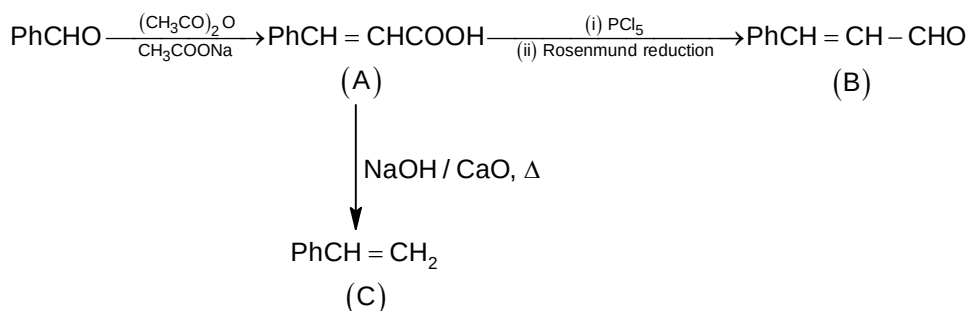
Ans. 20.80

36. If 0.1 mole of PhCHO is taken then mass of B formed (with 100 efficiency of the reaction)

Ans. 13.20

Sol. 0.1 mole of PhCHO gives 0.1 mole of Ph – CH = CHCHO (B)

$$\therefore W_{\text{PhCHCHCHO}} = \frac{132}{10} = 13.20 \text{ gm}$$



Question Stem for Question Nos. 37 and 38

Question Stem

0.5 gm of an organic compound was Kjeldahlised and the ammonia evolved was absorbed in to 60 ml of 0.1 N sulphuric acid solution. The residual acid solution was diluted with distilled water and the volume was made up to 150 ml. 50 ml of this solution required 20 ml of N/20 NaOH solution for complete neutralization.

[Atomic mass of N = 14, H = 1]

37. Calculate the mass percentage of nitrogen in the compound.

Ans. 8.40

$$\text{Sol. Meq. of residual H}_2\text{SO}_4 = \left(20 \times \frac{1}{20}\right) \times \frac{150}{50} = 3$$

$$\text{Meq. of H}_2\text{SO}_4 \text{ taken} = 60 \times 0.1 = 6$$

$\therefore$ Meq. of H_2SO_4 consumed = Meq. of $\text{NH}_3 = 6 - 3 = 3$

$$\text{Percentage of nitrogen} = \frac{1.4 \times 3}{0.5} \times 100 = 8.40$$

38. If we use Duma's process then the volume of N_2 at STP will be V lit. Calculate 100 V.

Ans. 3.36
(Range : 3.35 – 3.37)

Sol. Moles of $\text{N}_2 = \frac{1}{2}$ moles of $\text{NH}_3 = \frac{3}{1000} \times \frac{1}{2}$
 $= 1.5 \times 10^{-3}$
 $\therefore$ vol of N_2 at STP = $22.4 \times 1.5 \times 10^{-3} = 0.0336$ lit.
 $\therefore 0.0336 \times 100 = 3.36$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. If $\sin^{-1}(2^x) + \cos^{-1}(x^2) = \frac{\pi}{2}$, then the number of positive solution(s) is/are k_1 and total number of solution(s) of equation is/are k_2 , then

- (A) $k_2 - k_1 = 0$
 (B) $k_1 + k_2 = 2$
 (C) $k_2 + k_1 = 1$
 (D) $3k_1 + 2k_2 = 2$

Ans. C, D

Sol. Sol(s)

Method-1:

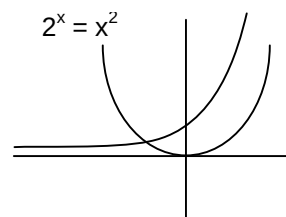
$$-1 \leq 2^x \leq 1 \Rightarrow x \in (-\infty, 0]$$

$$-1 \leq x^2 \leq 1 \Rightarrow x \in [-1, 1]$$

Domain $\in [-1, 0]$ so no positive solution can possible

Method-2:

No cut for $x \in \mathbb{R}^+$ so no positive real solution



40. Consider the two curves $C_1 : y^2 = 6x$ and $C_2 : x^2 + y^2 - 18x + 36 = 0$, then

- (A) C_1 and C_2 touches each other
 (B) C_1 and C_2 intersect each other at four distinct points
 (C) equation of common tangent of C_1 and C_2 is $x - 2y + 6 = 0$
 (D) equation of common chord of C_1 and C_2 is $x + y - 3 = 0$

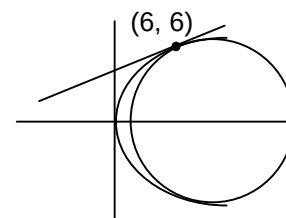
Ans. A, C

Sol. Equation of tangent to parabola at (6, 6) is $x - 2y + 6 = 0$

and this is tangent to given circle also

$$\text{Family of circle } (x - 6)^2 + (y - 6)^2 + \lambda(x - 2y + 6) = 0$$

Put $\lambda = -6$ we will get desired circle



41. If the equation $7 \tan^{-1}(3x^2 + 2x + c) + 3 \cot^{-1}(3x^2 + 2x + c) = \frac{5\pi}{2}$ has two distinct real solutions, then the range of c is super set of

(A) $(-\infty, -1)$

(B) $\left(-\infty, \frac{7}{3}\right)$

(C) $\left(-5, \frac{4}{3}\right)$

(D) $\left[\frac{4}{3}, 5\right)$

Ans. A, C

Sol. $7 \tan^{-1}(3x^2 + 2x + c) + 3 \cot^{-1}(3x^2 + 2x + c) = \frac{5\pi}{2}$

$$\Rightarrow \tan^{-1}(3x^2 + 2x + c) = \frac{\pi}{4}$$

$$\Rightarrow 3x^2 + 2x + c = 1$$

$$\Rightarrow 3x^2 + 2x + (c - 1) = 0 \text{ equation has two distinct real solution when } c < \frac{4}{3}$$

42. Let ABC be a triangle with sides a, b, c with corresponding angle α, β, γ respectively. The circum-radius of triangle is R . If $\alpha = 3\beta$, then $(a^2 - b^2)(a - b)$ is equal to

(A) bc^2

(B) b^2c

(C) $8R^3 \sin \beta \sin^2 \gamma$

(D) $8R^3 \sin^2 \beta \sin \gamma$

Ans. A, C

Sol. $(a^2 - b^2)(a - b) = 8R^3(\sin^2 \alpha - \sin^2 \beta)(\sin \alpha - \sin \beta)$
 $= 8R^3\{\sin^2 3\beta - \sin^2 \beta\} \{\sin 3\beta - \sin \beta\}$
 $= 8R^3\{\sin 4\beta \sin 2\beta \cdot 2 \cos 2\beta \sin \beta\}$
 $= 8R^3 \sin^2 4\beta \sin \beta$
 $= 8R^3 \sin^2(\pi - 4\beta) \sin \beta$
 $= 8R^3 \sin^2 \gamma \sin \beta$
 $= bc^2$

43. Let $p = \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$, where, A, B, C are angles of a triangle such that $2x^2 - 8x - 8p + 9 = 0$ for some real x , then

(A) $\sin A + \sin B + \sin C < \cos A + \cos B + \cos C$

$$(B) \quad \tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} < 1$$

$$(C) \quad \sin A + \sin B + \sin C > 1$$

$$(D) \quad \cos A + \sqrt{3} \sin B > \frac{3}{2}$$

Ans. C, D

$$\text{Sol.} \quad 64 - 8(-8p + 9) \geq 0 \Rightarrow p \geq \frac{1}{8}$$

$$\Rightarrow \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \geq \frac{1}{8}$$

$$\therefore \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{1}{8} \left[\because \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8} \right]$$

$$\Rightarrow A = B = C = \frac{\pi}{3}$$

44. In a given plane, let K and k be circles with radii R and r respectively and suppose that K and k intersect in precisely two points S and T . Let the tangent to k through S intersect K also in B and suppose B lies on the common tangent to K and k .

If ϕ is the interior angle between the tangents of K and k at S , then $\frac{r}{R} = \left(\lambda_1 \sin \frac{\phi}{2} \right)^{\lambda_2}$, then

$$(A) \quad \lambda_1 = 2$$

$$(B) \quad \lambda_2 = 2$$

$$(C) \quad \lambda_1 = 3$$

$$(D) \quad \lambda_2 = 3$$

Ans. A, B

Sol. Let common tangent contact k at A .
Let t denotes the length of AB and O and M are centres of k and K respectively
Now, $\angle MSB = 90^\circ - \phi = \angle MBS$ ($\triangle BSM$ is isosceles)
therefore $\angle ABS = \phi$ ($90^\circ - \angle MBS$)
also since BA and BS are tangents to k from a common point B

$$\Rightarrow BS = BA = t; \angle ABO = \angle OBS = \frac{\phi}{2}$$

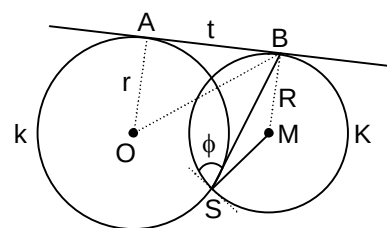
$$\text{In } \triangle ABO, \frac{r}{t} = \tan \frac{\phi}{2} \quad \dots (1)$$

In $\triangle BSM$ by cosine rule

$$R^2 = t^2 + R^2 - 2Rt \cos(90^\circ - \phi)$$

$$t = 2R \sin \phi \quad \dots (2)$$

From equation (1) and (2), we get



$$\tan \frac{\phi}{2} = \frac{\sin \frac{\phi}{2}}{\cos \frac{\phi}{2}} = \frac{r}{t} = \frac{r}{2R \sin \phi} = \frac{r}{4R \sin \frac{\phi}{2} \cos \frac{\phi}{2}} \Rightarrow \frac{r}{R} = 4 \sin^2 \frac{\phi}{2}$$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

Let $P(\alpha, \beta)$ be a point in the first quadrant. Circles are drawn through P touching the coordinate axes

45. Radius of one of the circles through P is

- (A) $(\sqrt{\alpha} - \sqrt{\beta})^2$
- (B) $(\sqrt{\alpha} + \sqrt{\beta})^2$
- (C) $\alpha + \beta - \sqrt{\alpha\beta}$
- (D) $\alpha + \beta - \sqrt{2\alpha\beta}$

Ans. D

46. Relation between α and β for which two circles are orthogonal is

- (A) $\alpha^2 + \beta^2 = 4\alpha\beta$
- (B) $\alpha^2 + \beta^2 = 6\alpha\beta$
- (C) $\alpha^2 + \beta^2 = \alpha\beta$
- (D) $\alpha^2 + \beta^2 = 2\alpha\beta$

Ans. A

Sol. (Q.45.-46)

Let circle be $(x - r)^2 + (y - r)^2 = r^2$

$$\Rightarrow r^2 - 2r(\alpha + \beta) + \alpha^2 + \beta^2 = 0$$

$$\Rightarrow r = (\alpha + \beta) \pm \sqrt{2\alpha\beta}$$

For orthogonal intersection

$$\Rightarrow r_1^2 + r_2^2 = 4r_1r_2 \Rightarrow \alpha^2 + \beta^2 = 4\alpha\beta$$

Paragraph for Question Nos. 47 and 48

Consider the curves $C_1 : y = \frac{8}{27}x^3$ and $C_2 : y = (x + a)^2$

47. The range of "a" for which there exists two common tangents to the curves C_1 and C_2 , other than x-axis is

- (A) $(0, \infty)$
 (B) $\left(-\frac{1}{2}, \infty\right) \sim \{0\}$
 (C) $\left(-\frac{3}{2}, \infty\right) \sim \{0\}$
 (D) $[1, \infty)$

Ans. B

48. If $a = 4$, then the area bounded by the two tangent lines mentioned in the previous question and the given parabola C_2 is

- (A) 100 sq. units
 (B) 144 sq. units
 (C) 180 sq. units
 (D) 200 sq. units

Ans. B

Sol. (Q.47-48)

Let (x_1, y_1) is contact point of common tangent on curve $C_1 : y = \frac{8}{27}x^3$ and slope of common tangent be m

$$\Rightarrow m = \frac{8}{9}x_1^2 \text{ and } 2x_1^2 - 6x_1 - 9a = 0$$

$$\Rightarrow a \in \left(-\frac{1}{2}, \infty\right) \sim \{0\} . \text{ If } a = 4, \text{ then } C_2 : y = (x + 4)^2$$

$$\Rightarrow \text{Common tangents are } y = 32x - 128 \text{ and } y = 8x + 16$$

$$\Rightarrow \text{Required area} = 144 \text{ sq. units}$$

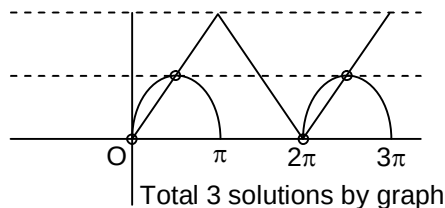
Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. The number of roots of equation $\sqrt{\sin x} = \cos^{-1}(\cos(x))$ in $(0, 3\pi)$ is

Ans. 3

Sol. Shown in the given figure



50. If the sum of infinite series

$$1 + k \cos \theta + k^2 \cos 2\theta + k^3 \cos 3\theta + \dots \infty \text{ is } L,$$

then max value of L is equal to (where $k = \frac{1}{2}$)

Ans. 2

Sol. Let $C = 1 + k \cos \theta + k^2 \cos 2\theta + k^3 \cos 3\theta + \dots \infty$

Let consider $S = k \sin \theta + k^2 \sin 2\theta + k^3 \sin 3\theta + \dots \infty$

$$C + iS = 1 + k(e^{i\theta}) + k^2(e^{i\theta})^2 + k^3(e^{i\theta})^3 + \dots \infty$$

$$= \frac{1}{1 - k \cdot e^{i\theta}} = \frac{1}{1 - k(\cos \theta + i \sin \theta)} = \frac{1 - k \cos \theta + ik \sin \theta}{1 + k^2 - 2k \cos \theta}$$

$$\Rightarrow C = \frac{1 - k \cos \theta}{1 + k^2 - 2k \cos \theta} = L \text{ (given)}$$

$$\text{Now, if } k = \frac{1}{2}$$

$$L = \frac{1 - \frac{1}{2} \cos \theta}{1 + \frac{1}{4} - \cos \theta} \Rightarrow \cos \theta = \frac{5L - 4}{4L - 2} \Rightarrow -1 \leq \frac{5L - 4}{4L - 2} \leq +1$$

$$L \in \left[\frac{2}{3}, 2 \right]$$

51. For each natural number $n \geq 2$, the largest possible value of the expression $V_n = \sin x_1 \cos x_2 + \sin x_2 \cos x_3 + \dots + \sin x_n \cos x_1$ is $f(n)$, then $f(100)$ is equal to (where $x_1, x_2, x_3, \dots, x_n$ are arbitrary real numbers)

Ans. 50

Sol. By the inequality $2ab \leq a^2 + b^2$, we get

$$V_n \leq \frac{\sin^2 x_1 + \cos^2 x_2}{2} + \frac{\sin^2 x_2 + \cos^2 x_3}{2} + \dots + \frac{\sin^2 x_n + \cos^2 x_1}{2} = \frac{n}{2}$$

with inequality for $x_1 = x_2 = \dots = x_n = \frac{\pi}{4}$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

A circle $S = 0$ has radius 10 units. Point P is outside the circle and shortest distance of P from circle $S = 0$ is 6 units. Chord RQ is parallel to OP, where O is centre of circle $S = 0$ and distance between RQ and OP is 3 units. If $(PR)^2 + (PQ)^2$ is a three digit number $\alpha\beta\gamma$, then

52. The value of $\alpha + \beta$ is equal to _____

Ans. 8.00

53. The value of $\gamma - \beta$ is equal to _____

Ans. 1.00

Sol. (Q.52-53)

$$(PQ)^2 = 9 + (16 - \sqrt{91})^2$$

$$(PR)^2 = 9 + (16 + \sqrt{91})^2$$

$$\Rightarrow (PQ)^2 + (PR)^2 = 712$$

$$\Rightarrow \alpha = 7, \beta = 1, \gamma = 2$$

Question Stem for Question Nos. 54 and 55

Question Stem

In a $\triangle ABC$, the three altitudes from vertices are along the lines $4x = y$, $x = y$, $x + 2y = 0$ and $A \equiv (\alpha, \alpha)$, where α is a parameter. If centroid of triangle is (α_1, β_1) and circumcentre of triangle is (α_2, β_2) , then

54. The value of $\alpha_1 + 3\beta_1 + 2$ is equal to _____

Ans. 2.00

55. The value of $2\alpha_2 + 6\beta_2 + 5$ is equal to _____

Ans. 5.00

Sol. Locus of centroid and circumcentre is $x + 3y = 0$

$$\Rightarrow \alpha_1 + 3\beta_1 = 0, \alpha_2 + 3\beta_2 = 0$$

Question Stem for Question Nos. 56 and 57**Question Stem**

Perpendicular is drawn from a fixed point (3, 4) to a variable line which cuts positive x-axis at one unit distance from origin. Circle $S(x, y) = 0$ represents the locus of the foot of perpendicular drawn from point (3, 4) to the variable line. If radius of $S(x, y) = 0$ is $\sqrt{\lambda_1}$ and length of tangent drawn to $S(x, y) = 0$ from origin is $\sqrt{\lambda_2}$, then

56. The value of λ_1 is equal to _____

Ans. 5.00

57. The value of λ_2 is equal to _____

Ans. 3.00

Sol. (Q.56-57)

Let $x + \lambda y - 1 = 0$ be the variable line and (h, k) be the foot of perpendicular on it from (3, 4)

$$\Rightarrow (h - 2)^2 + (k - 2)^2 = 5$$

$$\Rightarrow \text{Radius of } S(x, y) = 0 \text{ is } \sqrt{5}$$

$$\Rightarrow \text{Length of tangent from origin to } S(x, y) = 0 \text{ is } \sqrt{3}$$